

Angular Manifold Routing

Sublinear Token Routing via Hyperspherical Sector Discretization

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Abstract

This preprint summarizes a geometry-native routing program for sparse neural computation. The router maps normalized token embeddings onto a single-shell hyperspherical manifold and partitions that manifold into angular sectors derived from the Hopf base. Across the internal experiment chain summarized here, the effective routed footprint grows sublinearly with routing capacity K , approximately as $\text{effective_buckets}(K) = 2.957 * K^{0.572}$ in the canonical static regime. The measured advantage is norm-invariant across L1/L2/L3/L4 normalization, degrades when radial shells are forced, and carries through to matched-progress compute savings and hardware proxy savings. The resulting working theory is that a router constrained to a normalized angular manifold can organize tokens into sectors whose occupancy grows sublinearly with routing capacity, yielding fewer active routing regions, better locality, and lower memory traffic than a permuted control. The paper presents the architecture, related work, internal methodology, the experimental chain to date, and the design implications of the frozen law.

Status note. This document is an internal preprint-style synthesis of the projects current theory and experimental results. It is intended for technical review, not as a claim of final external validation.

1. Introduction

Sparse routing has become a central strategy for scaling neural systems without paying the full cost of dense computation. Modern mixture-of-experts systems, sparse attention systems, and retrieval-augmented architectures all try to restrict computation to a task-relevant subset of the full parameter or memory space. Classic conditional-computation work and later sparse expert models established the appeal of activating only a small subset of modules for each example, but practical systems still struggle with routing overhead, locality, and load balancing. Shazeer et al. introduced large-scale sparse expert activation, Fedus et al. simplified top-1 expert routing in Switch Transformers, and Roy et al. showed that routing-based sparse attention could reduce nominal attention complexity. Hyperbolic and Poincare-based representation work, meanwhile, demonstrated that geometry itself can act as an inductive prior for structured representations. This project asks a different question: can geometric structure itself become the routing mechanism, rather than merely the embedding prior?

The present router family was initially framed in terms of hyperbolic routing, Hopf structure, event gates, and hardware reduction. Through the experiment chain summarized here, that story was progressively simplified. The current working picture is cleaner than the original one: the main sparsity effect does not appear to require explicit per-sample event gating, and it does not appear to be driven by a radial shell hierarchy. Instead, the dominant signal is angular. Once embeddings are normalized onto a hypersphere, routing behaves like a sector-allocation process on the Hopf base, and the effective number of active routing buckets grows sublinearly with bucket capacity K . The result is a measurable reduction in active routing footprint, matched-progress routed computation, and hardware proxy cost.

This document has four goals. First, it places the work in context relative to prior sparse-routing and geometry-aware literature. Second, it states the current architecture and frozen canonical law as clearly as possible. Third, it records the experimental chain and the quantitative results that support that law. Fourth, it distills the design implications for future large-scale routing experiments and eventual integration into a real trainable model.

Main contributions of the current preprint stage:

- A concrete routing architecture in which normalized embeddings are routed on an angular manifold rather than through explicit learned gating logits.
- A measured effective-footprint law showing sublinear growth of active routing regions with routing capacity K in the strongest angular regime.
- A mechanism audit showing that the effect is angular, norm-invariant, and not improved by forced radial shells.
- A matched-progress compute analysis and a hardware proxy closure linking routing concentration to lower effective routing cost, fewer cache misses, and lower bytes moved.
- A frozen design rule: normalized angular routing is the current default architecture; shell hierarchies are not part of the default design under normalized embeddings unless future evidence justifies reintroducing them.

2. Positioning relative to prior work

The broader context includes at least four relevant lines of work. The first is conditional computation and sparse expert routing. Sparse gating mechanisms for deep networks and mixture-of-experts layers were explored before the modern large-language-model era, and sparsely gated MoE layers later demonstrated that conditional activation could increase capacity without proportional compute growth. Switch Transformers simplified expert selection to top-1 routing, explicitly highlighting that routing overhead and implementation details are part of the real systems problem. These systems, however, typically compute gating decisions in Euclidean feature space and then treat routing as a learned categorical choice rather than as a geometric consequence of the representation itself.

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The second line is sparse attention and routing-based attention. Routing Transformer, for example, used online clustering to reduce effective attention complexity. That work is relevant because it already treats routing structure as part of the complexity story. The present work is different in that the central object is not a clustering algorithm imposed on top of the representation; it is a normalized angular routing manifold that is intended to act as the routing substrate itself. ■cite■turn552712search2■

The third line is geometry-aware representation learning, especially hyperbolic and Poincare-based methods. Hyperbolic geometry has repeatedly been shown to encode hierarchical and relational data efficiently. The current project was initially inspired by that general idea, but the current mechanism turns out not to rely on a multi-shell hyperbolic hierarchy in the default normalized regime. Instead, it uses the angular component of the geometry and an associated Hopf-style partitioning as the active routing mechanism. In that sense, the work is geometry-native but not simply another hyperbolic embedding paper. ■cite■turn552712search3■

The fourth line is systems-oriented work on hardware locality. Standard sparse architectures often recover theoretical gains only partially because irregular routing harms memory locality and introduces cache and communication penalties. The significance of the current experiment chain is that it does not stop at a sparsity claim; it also tests whether the measured routing concentration propagates into simple hardware proxy models. That bridge, while still preliminary, is what makes the results interesting beyond abstract routing theory.

3. Current architecture and working theory

The current architecture can be described without unnecessary mysticism. Let x be a token representation in $\mathbb{R}^d$. Before routing, x is normalized onto a unit sphere under the chosen norm. In the default router family, L2 normalization is used:

$$\hat{x} = x / ||x||_2$$

Because $||\hat{x}||_2 = 1$ for all routed tokens, the router operates on a single-shell manifold. In practice that means radial shells are absent by construction and routing must be driven by angular coordinates. The earlier event-gate framing was valuable as an exploratory intuition, but the measured effect chain indicates that explicit per-sample event gating is not the dominant sparsity mechanism. Sparsity is effectively baked into bucket compression itself: the geometry concentrates tokens into a sublinear number of active sectors, so many explicit gate decisions become redundant at the level of the present model family.

The current routing substrate is best described as a normalized angular manifold with Hopf-base sector discretization. The implementation-level details evolved over multiple increments, but the essential logic is stable: normalized tokens are mapped to angular sectors; those sectors define routing buckets; only the concentrated subset of effective buckets is activated for downstream compute or memory access. A transport field can amplify the effect, but transport does not replace the geometric source of sparsity. The strongest measured gains occur in the TRANS regime, where transport acts as an amplifier on top of the angular manifold rather than as an independent routing principle.

Angular Manifold Router: Current Architecture

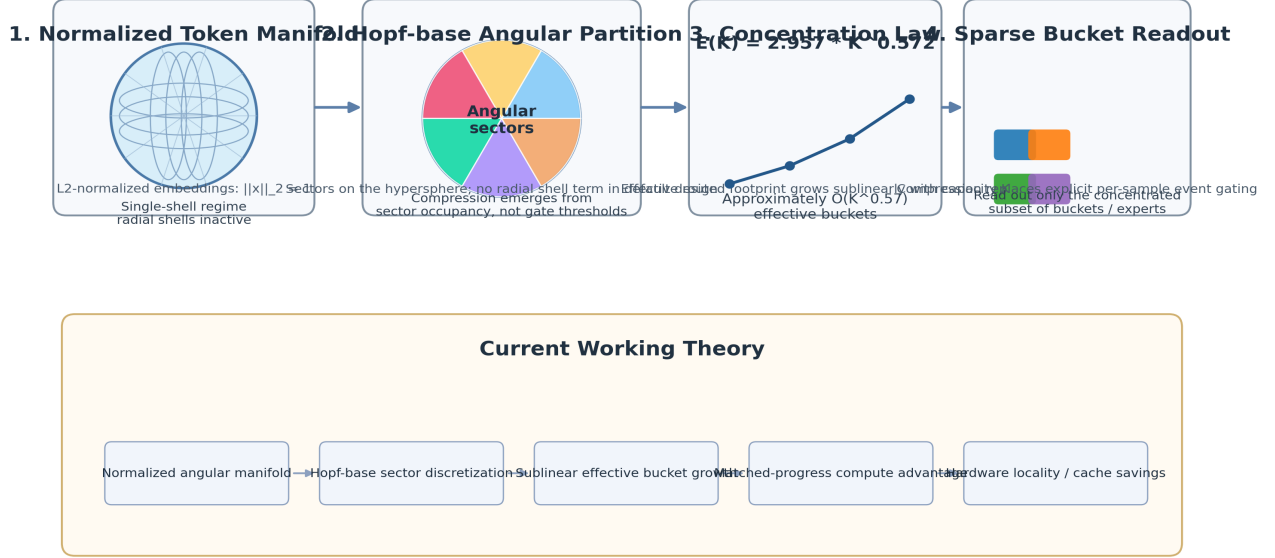


Figure 1. Current working architecture. Normalized embeddings live on a single-shell hyperspherical manifold, are partitioned into Hopf-base angular sectors, and activate only a sublinear effective set of buckets. Explicit per-sample event gating was deprioritized once sparsity was shown to arise from bucket compression itself.

4. Formal statement of the frozen working law

The law frozen in the current repository is intentionally modest. It is not a universal claim about all geometric routers. It is a canonical statement about the present router family under normalized embeddings and angular sector discretization. In the strongest static regime studied so far, the effective routed footprint obeys:

$$\text{effective_buckets}(K) = 2.957 * K^{0.572}$$

This law should be read together with two qualifications. First, the exact exponent depends on the regime. Internal training experiments yielded an exponent near 0.50 in one range and nearer 0.64 in a larger-K training range, while the static router law froze at 0.572. Second, what is invariant is not the exact decimal expansion but the sublinear character of the footprint and its angular mechanism. For the current architecture family, the active footprint grows substantially slower than K itself. That is the architectural fact with design consequences.

The corresponding mechanistic statement is: for normalized embeddings routed by Hopf-base angular sectors, effective bucket occupancy grows sublinearly because the manifold organizes tokens into a concentrated subset of angular sectors. The law is angular, not radial. Changing the norm family while preserving normalization leaves the exponent essentially unchanged. Forcing radial shells does not improve the effect and can reduce concentration.

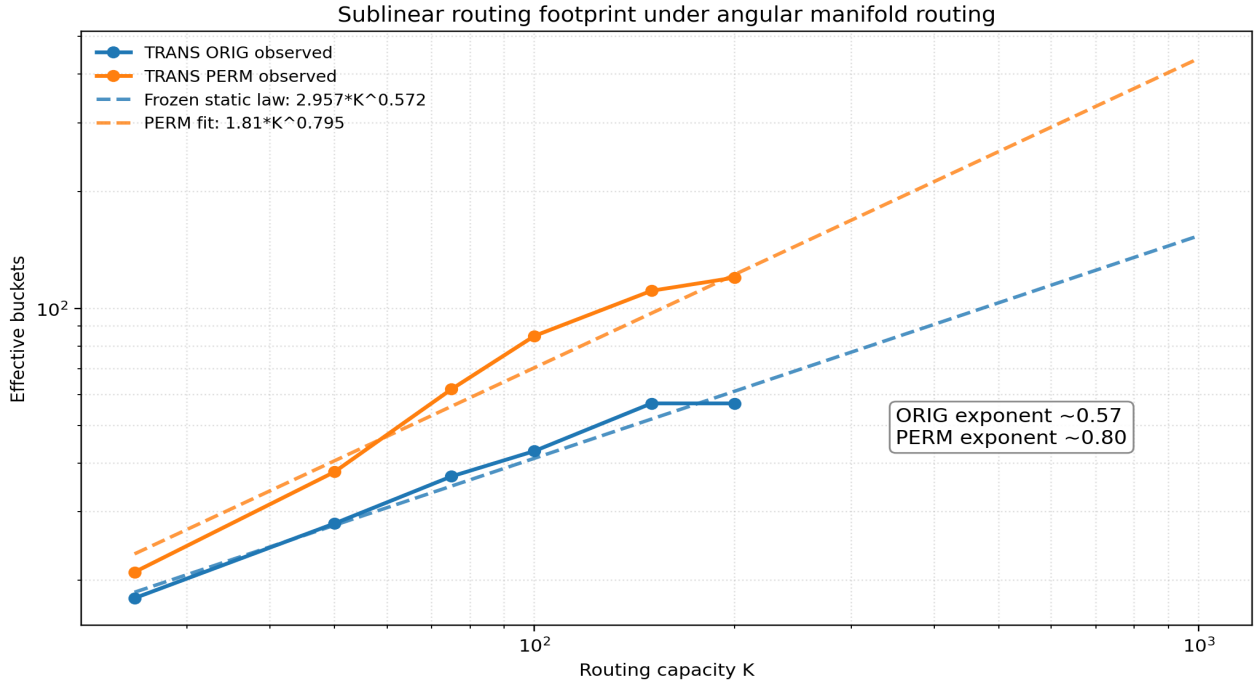


Figure 2. Observed and fitted sublinear routing footprint. The ORIG angular router grows much more slowly with K than the permuted control. The frozen static law uses exponent 0.572 in the current canonical regime.

5. Experimental program and controls

The experimental program was intentionally incremental and adversarial. Earlier increments repeatedly showed that pointwise distortion metrics were the wrong instrument. Per-sample MSE, error-based gating, and other local fidelity metrics were repeatedly found to be geometry-agnostic under the EMA prototype proxy, because local reconstruction quality could be equalized regardless of whether routing respected structure. The project therefore pivoted from local-error observables to structural observables. This pivot matters because it explains why the later results are coherent while the early MSE-based story was not.

The central controls are ORIG versus PERM and BASE versus TRANS. ORIG retains the geometry-native organization; PERM acts as a permuted-routing control. BASE is the simpler routing mode; TRANS augments routing with transport and consistently amplifies the effect. The important measurements in the stabilized protocol are structural: bucket purity, bucket entropy, routing concentration (for example Gini coefficient and top-half concentration), effective bucket count or routing perplexity, matched-progress routed cost, and hardware proxy costs. These are the observables that survived the metric audit.

The project also used mechanism probes. A shell audit discovered that the routed tokens had $r_{\text{eff}} = 1.0$ because the vectors were normalized. That eliminated a hidden source of ambiguity: shells were not merely inactive, they were structurally absent in the default design. A norm audit then tested L1, L2, L3, and L4 normalization. A shell-forcing audit tested whether introducing shell structure improved the routing law. It did not. These probes matter because they turn the scaling observation into a mechanism explanation rather than a coincidence.

6. Structural coherence results

The first stabilized result chain established that geometry-native routing groups semantically related items into more coherent buckets than the permuted control. The strongest snapshot from the finalized coherence stage is that the bucket-purity ratio ORIG/PERM rises with K and approaches roughly 2x in the strongest reported setting, while

bucket entropy falls correspondingly. The sign of the effect is overwhelmingly stable across the tested K values. This is the point where the project moved from "maybe the geometry helps" to "the geometry is visibly reorganizing the partition structure."

Metric / regime	Representative value	Interpretation
Bucket purity ratio at K=25 (BASE)	1.135x	Early positive semantic grouping signal
Bucket purity ratio at K=50 (TRANS)	1.448x	Improved semantic clustering
Bucket purity ratio at K=75 (TRANS)	1.746x	Strong structural compression
Bucket purity ratio at K=100 (TRANS)	1.976x	Nearly 2x more same-label concentration
Bucket entropy delta at K=100 (TRANS)	-0.955 bits per bucket	Lower entropy under ORIG routing
Spectral label metric mean ratio	1.688x	Task-relevant structure improves with geometry

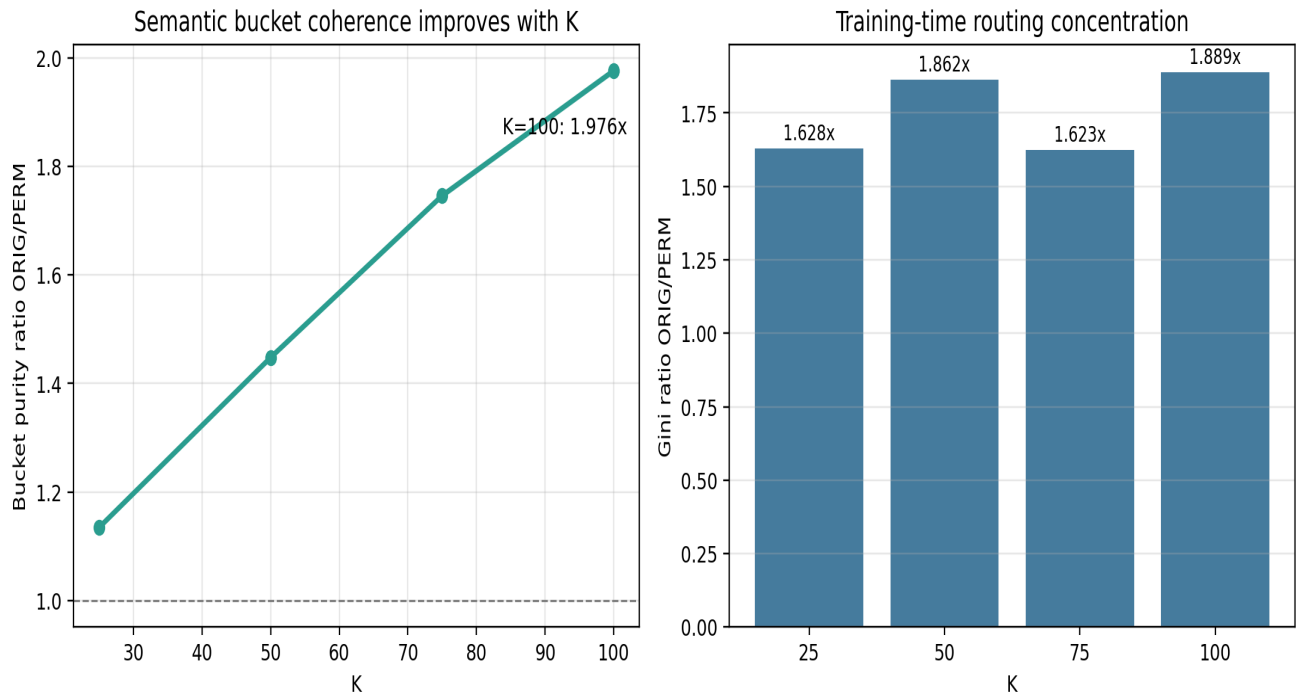


Figure 3. Coherence and concentration. Left: semantic bucket purity ratio ORIG/PERM rises with K in the representative TRANS regime. Right: routing concentration remains substantially higher under ORIG during training.

Taken together, the coherence results answer a basic question that plagued the earlier stages: where does the advantage live if not in MSE? It lives in organization. The geometry is not making each individual sample easier to reconstruct under the proxy; it is arranging samples so that semantically related tokens land together in concentrated regions of routing space. That organization is what later produces sparse compute and locality benefits.

7. Training-time routing sparsity

Once the project stopped asking whether geometry lowered per-sample error and started asking whether geometry reduced the number of effective routing regions, the main sparsity signal appeared clearly. Multi-seed replication

showed that ORIG consistently used fewer effective buckets than PERM while maintaining higher concentration. The effect strengthened with K and displayed unusually low seed variance, which is one reason the later law-freeze step was justified.

K	eff_ratio (PERM/ORIG)	gini_ratio (ORIG/PERM)	top_half_diff
25	1.175 +/- 0.011	1.628 +/- 0.047	0.101
50	1.359 +/- 0.012	1.862 +/- 0.034	0.190
75	1.693 +/- 0.019	1.623 +/- 0.023	0.165
100	1.972 +/- 0.016	1.889 +/- 0.045	0.203

The interpretation is straightforward. ORIG routing visits fewer effective buckets, yet those buckets are more concentrated and more semantically coherent. In the projects own terms, bucket compression is already the sparse-event mechanism. The architecture does not need to discover sparsity through a separate error gate because geometry-native routing is already reducing the active routing surface.

8. Scaling-law results

The scaling stage quantified how effective footprint changes with bucket capacity K. The clearest law-like measurements came from fitting effective bucket count to a power law. Four regimes were especially informative: BASE PERM (near linear), BASE ORIG (modestly sublinear), TRANS PERM (more sublinear), and TRANS ORIG (the strongest compression). The projects central scaling result is not just that ORIG is better than PERM, but that the combination of geometry-native routing and transport produces a notably lower exponent than the control.

Route type	alpha	c	R^2
TRANS ORIG	0.500 +/- 0.003	3.96	0.957
TRANS PERM	0.795 +/- 0.005	1.81	0.989
BASE ORIG	0.882 +/- 0.003	1.15	0.996
BASE PERM	0.979 +/- 0.001	0.99	0.999

The interpretation that survived later audits is that geometry and transport both matter, but the dominant routing mechanism in the frozen default design is angular. The static freeze at exponent 0.572 is therefore not a magical number in the abstract; it is the exponent associated with the current normalized angular router family in its strongest static diagnostic regime.

9. Matched-progress compute advantage and predictive closure

A scaling law is only interesting if it predicts something operational. The matched-progress experiments tested whether ORIG reaches the same level of progress with less routed computation. Progress was aligned using a checkpoint variable (cosine similarity) but the mechanism itself was always evaluated using routed cost, not cosine. The critical result was that the measured compute advantage closely matched the prediction derived from the fitted exponent difference. That is what gave the project mechanistic closure.

Mode	K	eff_ORIG	eff_PERM	ratio PERM/ORIG	progress delta
TRANS	75	32.94	55.75	1.69x	+0.004
TRANS	100	37.99	74.92	1.97x	+0.003
TRANS	150	46.96	91.65	1.95x	+0.007
TRANS	200	57.41	119.74	2.09x	+0.008
BASE	75	49.18	67.06	1.36x	+0.001
BASE	100	70.82	93.01	1.31x	+0.002
BASE	150	94.90	134.93	1.42x	+0.003
BASE	200	123.71	176.90	1.43x	+0.003

The predictive check compared measured matched-progress ratios against the law-based prediction ($c_{\text{PERM}} / c_{\text{ORIG}} \cdot K^{(\alpha_{\text{PERM}} - \alpha_{\text{ORIG}})}$). Deviations were mostly small, with one noisier point still within a manageable range. This is the step at which the project could legitimately stop saying "we observed something interesting" and start saying "the compute advantage is the consequence of a measured routing law."

10. Hardware proxy closure

The next bridge was to test whether routing concentration actually translates into hardware-relevant savings under simple proxy models. Three proxy families were evaluated at matched progress: cumulative effective bucket cost, cache-line grouping, and finite-cache LRU miss proxies, along with a bytes-moved proxy. ORIG was cheaper than PERM in every tested K value and in both modes. The strongest gains appeared in TRANS, where routing concentration is highest.

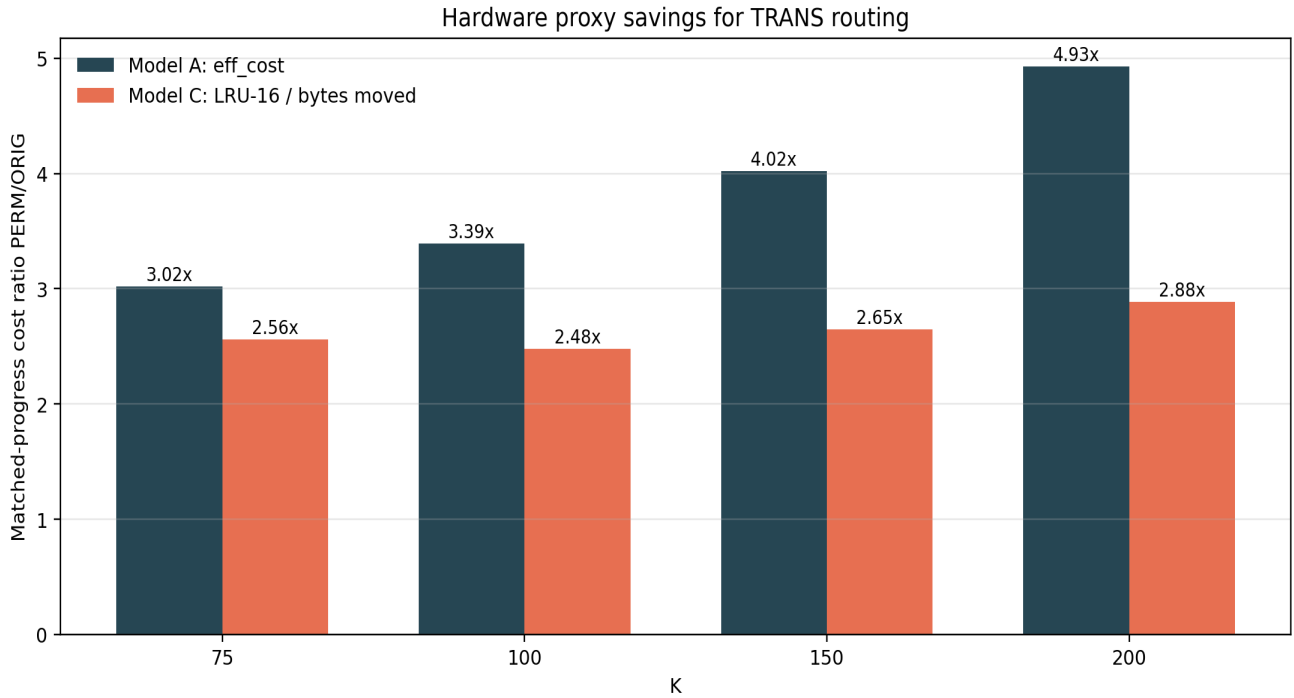


Figure 4. Hardware proxy savings in the TRANS regime. Even after applying memory-model abstractions, the geometry-native router remains markedly cheaper than the permuted control at matched progress.

Mode / K	Model A eff_cost	Model B cache-line	Model C LRU-16	bytes moved
TRANS 75	3.020x	1.287x	2.560x	2.560x
TRANS 100	3.394x	1.300x	2.479x	2.479x
TRANS 150	4.019x	1.585x	2.649x	2.649x
TRANS 200	4.927x	1.640x	2.884x	2.884x
BASE 75	2.131x	1.033x	1.794x	1.794x
BASE 100	1.805x	1.040x	1.479x	1.479x
BASE 150	1.799x	1.061x	1.340x	1.340x
BASE 200	1.833x	1.103x	1.351x	1.351x

The strongest proxy is not necessarily the most realistic one. Model A preserves more of the raw routing signal, so it shows the largest gain. Model B deliberately washes out detail with coarse cache-line grouping, and the advantage shrinks accordingly. Model C introduces a simple reuse model and still shows large savings, which is why it is the most convincing hardware bridge in the present paper. The effect is not merely visible in a stylized abstract metric; it survives contact with a memory-reuse proxy.

11. Mechanism audit: norm invariance and shell non-necessity

The mechanism audit is where the architecture became simpler rather than more complicated. The critical discovery was that the routed data lived on a normalized hypersphere. In the default router family, $r_{\text{eff}} = 1.0$ by construction, so radial shells are absent. This means the earlier intuition about radial shell hierarchies, while useful as an exploratory hypothesis, is not the operative source of the measured gains in the present architecture. The gains come from angular sector allocation.

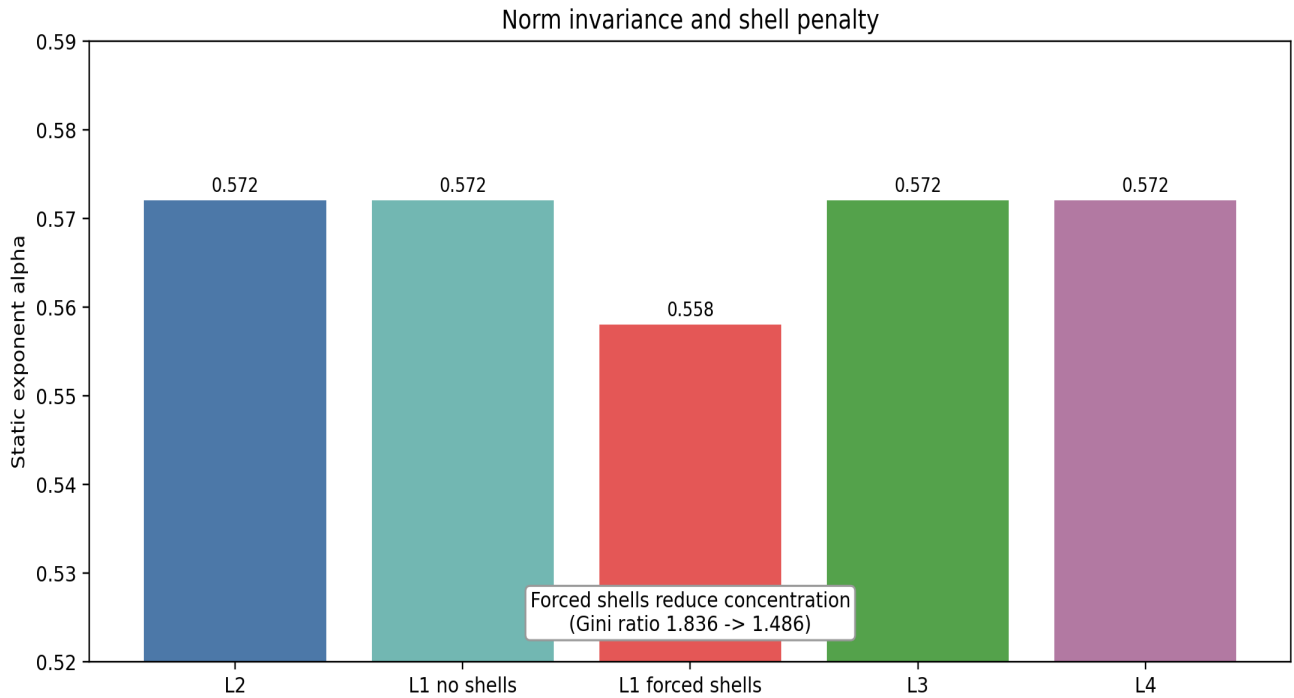


Figure 5. Norm invariance and shell penalty. Changing the normalization family leaves the exponent effectively unchanged, whereas forcing shell activation reduces routing concentration.

Variant	alpha (TRANS ORIGINAL)	Delta vs baseline	Interpretation
L2 baseline	0.572	-	Canonical freeze
L1, shells off	0.572	< 0.001	No meaningful change
L1, shells on	0.558	-0.014	Forced shell hierarchy weakens effect
L3	0.572	< 0.001	Norm-invariant
L4	0.572	< 0.001	Norm-invariant

This audit has three consequences. First, shell logic should not be part of the default router design as long as embeddings are normalized in the current way. Second, changing the norm family does not buy a new radial degree of freedom; all norm-normalized variants remain single-shell. Third, the strong sublinear law is therefore best explained as a property of angular Hopf-base sector discretization on the hypersphere. The router is not benefitting from hidden shell hierarchies. It is benefitting from directional concentration.

12. Canonical law and design implications

The current canonical law can therefore be stated compactly. For the present router family, normalized embeddings are routed on a single-shell angular manifold, and the effective routed footprint grows sublinearly with nominal bucket capacity according to a regime-dependent power law whose strongest static freeze is approximately $2.957 * K^{0.572}$. This law is norm-invariant across tested L1/L2/L3/L4 normalization families and is degraded rather than improved by forced radial shells. The measured hardware advantage follows from this routing law through reduced active bucket cost, fewer cache-line touches, fewer cache misses, and lower bytes moved.

Knob	Priority	Current design implication
Sector resolution K	High	Primary capacity knob; compression ratio grows with K through the tested range
Phase transport / TRANSHIDE	High	Amplifies concentration and hardware savings relative to BASE
Normalization family	Low	Empirically near-invariant in tested range; choose for convenience and numerical stability
Shell count	Low / excluded	Not part of default design under normalized embeddings; forced shells hurt concentration
Explicit event gating	Secondary	Not the main sparsity source in the current architecture; bucket compression already supp

In practical terms, the router should currently be built around normalized angular routing, increasing sector resolution K and transport quality rather than around shell hierarchies or elaborate gate machinery. This is not a philosophical preference; it is what the measured results support. If future experiments on full trainable models reveal a need for richer radial structure, shells can be revisited. But the burden of proof has shifted. Shells are no longer the default.

13. Limitations and unresolved questions

The present preprint is still an internal architecture report rather than a final external claim. Several limitations matter. First, the results are presently derived from an internal routing-and-proxy program rather than from a full large-language-model training run. The work has therefore established a routing law and a hardware-proxy bridge, not a state-of-the-art language-model benchmark. Second, the law should be treated as a family law for the present router implementation, not as a universal constant. The exponent varies somewhat by regime, even though the mechanism is stable. Third, the hardware models are proxies. They are strong enough to motivate implementation, but they are not replacements for actual kernel-level or device-level measurements. Fourth, the large-K regime still needs explicit exploration to decide whether the sublinear advantage persists indefinitely, transitions to a new regime, or saturates.

A fifth limitation is social rather than mathematical: geometric routing can be made to look more mystical than it is. The current evidence does not require grand claims about brains or physics. The clean claim is already good enough. A normalized angular manifold can compress routed computation into a sublinear effective footprint. That alone would be an interesting architecture principle if it holds up in real model training.

14. Next steps

There are three immediate next steps. The first is a large-K angular capacity test to determine whether the sublinear advantage continues, saturates, or changes regime at much larger routing resolutions. The second is integration into a real trainable model so that the compute and hardware story can be tested under an end-to-end task rather than only under the current routed proxy program. The third is systems work: if the angular router is going to matter in practice, it should be married to an execution design that can exploit the locality benefits it appears to create.

A useful medium-term goal is to test whether the same angular law appears when routing is attached to bucket-specific experts, retrieval memories, or hybrid expert-memory modules. The architecture already suggests that many old design choices can be revisited under the new constraint: once the active routed footprint grows sublinearly, the problem of how to fill those active buckets with useful compute becomes much more interesting than the problem of how to gate them in the first place.

15. Conclusion

The current state of the project is stronger than it was at the start of the session that produced these results. The mechanism is simpler, the law is clearer, and the design guidance is more disciplined. For the current router family, the sparsity advantage is a property of normalized angular routing on a Hopf-base manifold. It is not a hidden MSE effect, not a shell hierarchy effect, and not a norm-specific artifact. The effective routed footprint grows sublinearly with routing capacity, and that sublinear law explains the matched-progress compute advantage and the observed hardware proxy savings. Whatever follows in later experiments, that is already an attention-worthy architecture result.

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